

Software Development

Algorithms and Data Structures

IDENTIFICATION

CODE : IF-3-ALGO
ECTS : 2.0

HOURS

Lectures : 12.5 h
Seminars : 12.0 h
Laboratory : 0.0 h
Project : 0.0 h
Teacher-student
contact : 24.5 h
Personal work : 25.0 h
Total : 49.5 h

ASSESSMENT METHOD

Supervised test on the platform.

TEACHING AIDS

Lectures notes
Programming platform (based on
domjudge)

TEACHING LANGUAGE

French

CONTACT

M. GUERIN Eric
eric.guerin@insa-lyon.fr
M. SCUTURICI Vasile-Marian
marian.scuturici@insa-lyon.fr

AIMS

The objective of this module is to give algorithmic basics that are needed for any computer science engineer.
Students are initiated to the basic techniques for the design of a correct and good performing algorithm.
Classical algorithms are presented.

This module develops the skill "Implémenter de bons logiciels" together with the capacity "Choisir les algorithmes et structures de données adaptés à la situation, et évaluer leur complexité".

CONTENT

- A. Design and analysis of an algorithm
 - introduction + motivation
 - pseudocode: variable, expressions, assignment ; looping instructions and branching ; parameters, procedures/functions ;
 - complexity ;
 - correctness ;
 - preconditions/postrelations/invariants ;
- B. Algorithmic resolution paradigms : complete search, divide&conquer, greedy, dynamic programming
- C. Sorting algorithms
- D. Data structures
 - Dynamic allocation ;
 - Structures ;
 - Abstract data types ;
 - Stacks and queues ;
 - Vectors ;
 - Linked lists ;
 - Trees

BIBLIOGRAPHY

Steven S. Skiena, The Algorithm Design Manual, Springer, Second Edition, 2010

Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, Clifford Stein. Introduction to Algorithms, MIT Press, Third edition, 2009

Donald E. Knuth. Fundamental Algorithms, volume 1 of The Art of Computer Programming. Addison-Wesley, 1968. Third edition, 1997

UVa Online Judge, <http://uva.onlinejudge.org/>

DOMjudge - Programming Contest Jury System, <http://www.domjudge.org/>